Effective Projections from the Pre-Geometric Scalar Lattice:

Origins of Torsion, Dimensional Emergence, and Emergent Structures

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Abstract

We present a focused examination of effective projections from the pre-geometric scalar lattice of Entropic Universe Theory (EUT). The ontology consists of a bare countable index set equipped with a single scalar entropy density S_i and temperature-modulated affinities w_{ij}. Dirichlet energy minimization at the fixed working value $\beta \simeq 3$ produces emergent three-dimensional geometry.

Building on a continuum limit established via Γ-convergence of the discrete Dirichlet energy to the continuum kinetic term $\int \frac{1}{2}|\nabla S|^2\sqrt{\gamma}\, d^3x$, we show that an Einstein–Hilbert contribution arises and that diffeomorphism invariance emerges approximately at finite resolution and exactly in the continuum limit. Earlier attempts to fix the continuum prefactor by a classical isostatic count of the mean coordination number are withdrawn; the overall scale of the curvature term remains an open internal question. Recent structural measurements further withdraw an intermediate residual long-range layer interpretation of the elevated coordination.

The same projection logic is applied to torsion: the antisymmetric (directed) response of the lattice bonds under local entropy-gradient flow and temperature modulation is shown to coarse-grain to an effective vector potential A_μ whose continuum dynamics are those of a sourced Proca equation, together with the associated torsion tensor contribution. The "coiling" is thereby identified as a mathematical artifact of bond asymmetry rather than an independent geometric structure.

This treatment is shown to be consistent with the prior construction of black-hole interiors as chart-dependent partitions around Planck-scale 0D core knots of residual primordial bonds, with the observed Schwarzschild radius interpreted as a capture radius. Parallel arguments are given for the emergence of time from temporal stacking and for the overlapping of multiverse fibers at a shared coldest vertex. The Two-State Vector Formalism is treated as a non-fundamental, optional effective description rather than a required ingredient of the lattice ontology.

All equations and empirical predictions of the existing EUT framework remain unchanged. The paper supplies a unified conceptual and mathematical foundation for the continuum limit and for the status of torsion and related geometric structures as effective projections of the underlying scalar lattice.

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Foreword

This work did not begin as an attempt to construct a "theory of everything." When I started, I was not even aware that such an ambition occupied so many people.

My path began in artificial intelligence. While working with transformer-based models I noticed that certain physical and structural principles used to understand those systems appeared to reach further than the usual discussions allowed. At the same time, questions about the nature of existence that had troubled me for most of my life — what the universe is made of, how it holds together, what our place in it might be, and why there is something rather than nothing — remained unresolved. Those questions functioned, in the old phrase, like a splinter in the mind.

The framework that follows is the result of letting those two lines of thought inform each other. I do not claim it is final or complete. I do claim that its foundations are coherent enough, and its internal consistency strong enough, to justify continued examination.

I release this work freely and with clear intention: that it may be of some use to anyone else who is trying to find answers to the same questions. Knowledge of this kind is not meant to be locked behind closed doors. Whether the present formulation ultimately proves correct, incomplete, or wrong, the effort of thinking carefully about these matters is itself worthwhile.

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Plain-Language Summary

Imagine the universe starts not with space and time already in place, but with a collection of simple points, each carrying a single number that measures local "disorder" or entropy. These points are connected to one another by links whose strength depends on how different their entropy numbers are and on a temperature-like quantity. There is no built-in up, down, left, right, or even a built-in sense of distance—only the points and the links.

When the system settles into the arrangement that best balances the cost of sharp differences against the benefit of having many useful connections, something remarkable happens. Three-dimensional space emerges. A sense of time appears by stacking successive copies of the arrangement. Even the branching of possible histories and the structure of black-hole interiors can be understood as further consequences of the same settling process.

Many of the mathematical tools we use to describe the resulting universe—ordinary geometry, torsion (a kind of twisting of space), and so on—are not fundamental ingredients. They are effective descriptions: convenient ways of talking about the patterns that the underlying points and links produce. Just as we can talk about the "size" of a black hole's influence even though the true interior is a tiny, dimensionless knot of the original links, we can talk about torsion as a mathematical coiling that arises from directed differences in the links. The coiling itself is not a separate physical object; it is a useful description of an asymmetry that is already present in the connections.

This paper makes that projection idea precise. It shows how the discrete network of points and links becomes the continuous geometry of everyday physics, and it applies the same logic consistently to torsion, the emergence of dimensions, time, and other structures. Nothing in the existing predictions of the theory is changed. The goal is simply to make the foundations clearer and more unified.

1. Introduction and Motivation

Plain-language overview.
This paper starts from a simple idea: the universe begins as a network of points, each carrying a single number that measures local entropy, connected by links whose strength depends on temperature. No space, no time, and no extra fields are assumed at the outset. When the network settles into its lowest-energy arrangements, ordinary three-dimensional space, a directed sense of time, and even branching histories appear. The familiar mathematical structures we use to describe those features—metrics, torsion, successive moments, and so on—are not additional ingredients. They are simply the precise language we need in order to talk about patterns that the original network already produces. The goal of the paper is to make this projection idea clear and rigorous, without changing any of the theory's existing predictions.

The Entropic Universe Theory (EUT) begins with a deliberately minimal pre-geometric ontology: a bare countable index set $\{i\}$ carrying a single scalar entropy density S_i together with temperature-modulated affinities

$$w_{ij} = \left[1 + \frac{(S_i - S_j)^2}{T_i T_j}\right]^{-\beta}, \qquad \beta \simeq 3. \tag{1}$$

No intrinsic metric, no prior spacetime, and no additional fields are postulated. All observable structure—three spatial dimensions, an emergent metric, finite propagation speed, the arrow of time, black-hole entropy, and the low-entropy past—arises from free-energy minimization of the associated Dirichlet energy

$$E_{\text{spatial}} = \frac{1}{2}\sum_{i,j} w_{ij}(S_i - S_j)^2 \tag{2}$$

at the fixed working value $\beta \simeq 3$.

Once the continuum limit is taken, the same scalar dynamics are described by the effective field theory of the companion paper (EUT I). In that continuum language one encounters geometric structures—Riemannian metrics, torsion, connections—that do not exist as independent objects in the underlying lattice. They are *effective projections*: mathematical descriptions of patterns already present in the scalar affinities and residual primordial bonds.

Two concrete examples illustrate the pattern clearly. First, black-hole interiors are constructed as chart-dependent partitions around fixed Planck-scale 0D core knots of residual primordial bonds; the observed Schwarzschild radius $r_s = 2M$ is a capture radius in the exterior chart, not the size of an extended interior object. Second, the antisymmetric (directed) response of the lattice bonds under local entropy-gradient flow and temperature modulation projects, after coarse-graining, to an effective vector potential A_μ whose continuum dynamics are those of a sourced Proca equation, together with the associated torsion tensor contribution. In both cases a lower-dimensional or purely discrete feature is articulated using the richer language of the emergent continuum geometry.

The present note makes this projection principle precise and systematic. We first supply a continuum limit of the discrete Dirichlet energy via Γ-convergence, recovering the continuum kinetic term and an emergent Einstein–Hilbert contribution (whose overall scale is left as an

open internal question after withdrawal of the earlier classical isostatic matching and of an intermediate residual-layer interpretation), together with approximate diffeomorphism invariance. We then show that the identical projection logic accounts for the continuum description of torsion, for the emergence of temporal structure from stacking, and for the overlapping of multiverse fibers at a shared coldest vertex. The Two-State Vector Formalism is treated throughout as a non-fundamental, optional effective description.

No equations or empirical predictions of the existing EUT framework are altered. The goal is conceptual and mathematical clarification: to exhibit the continuum limit and the status of geometric structures as effective projections of a single scalar lattice, thereby strengthening the internal consistency of the theory.

2. Continuum Limit of the Discrete Dirichlet Energy

Plain-language overview.
The network of points and links described in the introduction is discrete. The familiar equations of continuum physics are continuous. This section shows, with mathematical care, how one passes from the discrete network to the continuous description without adding anything new. The energy that measures how uneven the entropy values are across the links is shown to converge to the ordinary continuum energy that appears in the effective field theory.

Along the way we find that the strength of gravity cannot be read off from a simple classical counting of links. Direct examination of the network at the working value $\beta \simeq 3$ shows a high-coordination, low-clustering geometry consistent with three-dimensional continuum behavior. The overall scale of the continuum curvature term is therefore left as an open internal question rather than being fixed by an imported classical formula. Everything that follows in the paper rests on this cleaned continuum bridge.

2.1. Setup and Scaling Limit

Let $\{i = 1, \ldots, N\}$ $(N \to \infty)$ be the bare index set equipped with entropy densities S_i and site-local temperatures $T_i > 0$. The bond affinities are

$$w_{ij} = \left[1 + \frac{(S_i - S_j)^2}{T_i T_j}\right]^{-\beta}, \qquad \beta \simeq 3, \tag{3}$$

and the spatial Dirichlet energy on any finite patch $\Lambda \subset \{i\}$ is

$$E_{\text{spatial}}(\{S_k\}_{k\in\Lambda}) = \frac{1}{2} \sum_{i,j\in\Lambda} w_{ij}(S_i - S_j)^2. \tag{4}$$

At each emergent time slice the lattice is partitioned into overlapping local observer charts p of fixed physical size L ($a \ll L \ll$ horizon). Inside chart p the spatial cutoff

$$w_{ij}^{(p,\text{spatial})} = w_{ij} \cdot \Theta\big(|S_i - S_j| < r_{\text{local}}^{(p)}\big) \cdot \Theta\big(w_{ij} > w_{\max}^{(p)} e^{-1}\big) \tag{5}$$

is applied, where $r_{\text{local}}^{(p)}$ is the patch correlation length. After minimization at the local working value $\beta_{\text{eff}}^{(p)} \approx 3$, the restricted graph is a weighted random geometric graph whose effective distance satisfies

$$d_{\text{eff}}(i,j)^2 \propto \frac{(S_i - S_j)^2}{T_i T_j}. \tag{6}$$

This distance is consistent with a Riemannian metric γ_{ab} on an emergent three-dimensional manifold (numerically verified by the correlation dimension converging to 3).

We consider the joint limit $a \to \ell_{\text{Pl}}$ (lattice spacing identified with the single Planck-scale ultraviolet cutoff) while keeping the physical chart size L fixed and letting the number of sites per chart $N_p \to \infty$. Global statements are obtained by summing over the atlas with controlled overlap errors given by the parallel-gradient transition functions.

2.2. Γ-Convergence of the Discrete Dirichlet Energy

After minimization at $\beta_{\text{eff}}^{(p)} \approx 3$, the weights $w_{ij}^{(p,\text{spatial})}$ are locally Lipschitz in the field differences and the graph is sparse with bounded degree. The discrete energy therefore takes the standard form of a weighted graph Dirichlet functional on a point cloud sampled from a Riemannian manifold.

After the Dirichlet energy has been minimized and the spatial cutoff applied, the surviving bonds are those for which $|S_i - S_j|$ remains modest. In any local chart the affinity then admits the expansion

$$w_{ij} \;\approx\; \Big(1 + \frac{(S_i - S_j)^2}{T_i T_j}\Big)^{-\beta} \;\simeq\; \exp\Big(-\beta \frac{(S_i - S_j)^2}{T_i T_j}\Big) \tag{7}$$

for the dominant (high-weight) edges. The quadratic form that appears in the discrete Dirichlet energy is therefore controlled by the local quantity

$$\frac{(S_i - S_j)^2}{T_i T_j} \;\sim\; \gamma^{ab}(x)\, \partial_a S\, \partial_b S. \tag{8}$$

The density of such bonds, once the continuum limit is taken, supplies the volume factor $\sqrt{\gamma}$. In this sense the affinity already encodes both the inverse metric that contracts the gradients and the measure that appears in the continuum integral. Once this identification is made, the standard Γ-convergence and spectral-convergence theorems for weighted geometric graphs apply directly to a discrete energy whose edge weights are those of a Riemannian Dirichlet form, yielding

$$E_{\text{spatial}}^{(p)} \;\xrightarrow{\Gamma}\; \int_p \tfrac{1}{2} |\nabla S|^2 \sqrt{\gamma}\, d^3x. \tag{9}$$

The argument above is a scaling identification, not a fully controlled derivation with explicit error estimates for these particular temperature-modulated weights. A tailored Γ-convergence proof remains a natural next mathematical step. The present identification is offered only to make transparent why the specific form of w_{ij} produces a continuum kinetic term of Riemannian type.

By the Γ-convergence results of García Trillos & Slepčev for Dirichlet forms on point clouds, and related results for geometric graphs, we have the stated continuum limit in the sense of

Γ-convergence in the Sobolev space $H^1(p)$ as $a \to 0$ with $N_p \to \infty$. The limit functional is taken with respect to the emergent metric γ_{ab} induced by d_{eff}.

The nonlinear dependence of the original w_{ij} on $|S_i - S_j|$ is handled variationally: minimization at the rigidity fixed point first produces an effective kernel that is locally equivalent to a rescaled neighborhood kernel on the emergent manifold; the standard theorems then apply directly to the post-minimization, cutoff graph. Global convergence on the whole emergent spacetime follows by summing the chart-wise energies and passing to the limit using the atlas transition functions (whose energy cost vanishes as $a \to 0$).

The same convergence implies spectral convergence of the associated graph Laplacian to the Laplace–Beltrami operator on the emergent manifold, with rates controlled by the sampling density and the regularity of the temperature field T.

2.3. Emergence of the Einstein–Hilbert Term

Once the discrete energy Γ-converges to the continuum Dirichlet integral, variation with respect to the emergent metric γ_{ab} yields the entropic stress-energy tensor

$$T^{\text{ent}}_{\mu\nu} = \partial_\mu S \, \partial_\nu S - \tfrac{1}{2} g_{\mu\nu} (\partial S)^2 \tag{10}$$

exactly as required by the continuum effective field theory.

The Einstein–Hilbert term is expected to arise because the quadratic form E_{spatial} encodes the integrated curvature of the bond network. In earlier formulations this connection was expressed by identifying the mean coordination number $\langle k \rangle$ with the prefactor of the Einstein–Hilbert action via a classical isostatic (Maxwell) count. That identification is no longer tenable.

Direct measurement of the surviving graph at the working value $\beta \simeq 3$ ($N \approx 16$k, corrected weight formula, improved high-weight sampling) shows:

- a robust correlation dimension $d_{\text{eff}} \approx 3$,
- a high mean degree ($\langle k \rangle \approx 30$, saturating the practical neighbor budget used in the present implementation),
- low average clustering ($C \approx 0.023$),
- essentially zero combinatorial tetrahedra (K_4).

The elevated coordination is carried by a dense population of strong bonds (mean weight ≈ 0.94, $> 99\,\%$ of surviving edges have $w > 0.7$). There is no significant population of weaker "residual" edges above the spatial cutoff. The graph is therefore better characterized as a high-coordination, low-clustering network consistent with an emergent three-dimensional continuum geometry, rather than as a sparse local geometry supplemented by a long-range residual layer.

Because the coordination statistics do not correspond to a classical rigid framework, the mean coordination number cannot be inserted into a Maxwell-style formula to fix the continuum value of G. The overall scale of the Einstein–Hilbert term therefore remains an open internal question: it must ultimately be expressed in terms of quantities native to the scalar lattice, not in terms of an imported isostatic count.

The continuum limit of the kinetic term and the existence of an emergent Einstein–Hilbert contribution are unaffected. Only the previous quantitative matching for the prefactor has been withdrawn. Supporting numerical scripts that reproduce the clustering, triangle counts, and weight distribution are deposited with this version of the paper.

2.4. Approximate Diffeomorphism Invariance

The atlas of local observer charts already supplies a discrete notion of coordinate freedom: overlapping charts are related by parallel-gradient offset transition functions. These functions are the precise discrete precursors of smooth coordinate changes on the emergent manifold.

Inside any fixed chart the converged graph Laplacian approximates the intrinsically coordinate-independent Laplace–Beltrami operator. A smooth diffeomorphism on the emergent 3-manifold can therefore be approximated by a smooth relabeling (or interpolation) of nearby indices. Because the post-cutoff energy is local, the energy cost of such a relabeling is $O(a)$. Consequently the discrete theory is approximately diffeomorphism invariant, with symmetry-breaking errors that vanish in the continuum limit $a \to 0$.

2.5. Numerical Support

The convergence statements are supported by existing large-scale simulations and by the supporting numerical scripts deposited with this version:

- embedded correlation dimension $d_{\text{eff}} \approx 3$ at $\beta \simeq 3$;
- positive algebraic connectivity of the patch Laplacian;
- geometric soft-kernel baseline yielding $\langle k \rangle \approx 3.3$–$3.4$ (demonstrating that the elevated coordination of the full graph is dynamical rather than a generic geometric consequence of three-dimensionality);
- on the actual surviving graph: mean degree ≈ 30, average clustering coefficient $C \approx 0.023$, and essentially zero combinatorial tetrahedra (the elevated coordination is carried by a dense population of strong bonds).

Future higher-resolution runs can quantify the rate at which the discrete energy approaches the continuum Dirichlet integral and the decay of diffeomorphism-breaking residuals.

With the continuum limit of the kinetic term established, every equation of the effective field theory follows from the single scalar line of the pre-geometric ontology. The same projection logic will now be applied to torsion and the remaining geometric structures.

3. Asymmetric Bond Responses and Effective Torsion

Plain-language overview.
Some of the links in the original network are not perfectly symmetric: the influence one way along a link can differ slightly from the influence the other way. This directed imbalance is real in the network, but it is not a new geometric object. When we describe the network with

continuum mathematics, the imbalance appears as torsion—a kind of twisting of space. The twisting is simply our mathematical way of talking about the directed imbalance that was already present among the links. A simple relational rule keeps every link positive overall, so the directed imbalances can never grow strong enough to tear the geometry apart or create causal paradoxes. The rest of this section makes that correspondence precise.

3.1. Lattice Origin of Directed Asymmetry

In the pre-geometric lattice the bond affinities w_{ij} are real and positive by construction. Under local entropy-gradient flow, however, the instantaneous rate of change $\partial_t(\nabla S)$ together with temperature modulation induces a small directed (antisymmetric) response in the effective stiffness. This response can be represented as a phase or imaginary component acquired by the bonds that participate in residual primordial connections—the same high-stiffness links that survive spatial cutoff and mediate entanglement and black-hole entropy counting.

The directed component is not an independent geometric structure. It is a mathematical encoding of the fact that entropy flow along a local gradient is not symmetric with flow against that gradient. In the language used earlier, the bonds acquire a mathematical "coiling" that shortens or stiffens them preferentially in one direction; the coiling itself exists only as a description of that asymmetry.

3.2. Coarse-Graining to an Effective Vector Potential

When the antisymmetric part of these phase gradients is coarse-grained over the residual primordial bonds and passed to the continuum limit, one obtains an effective vector potential

$$A_\mu \propto \left\langle \partial_{[\mu}(\nabla_{\nu]}S)/T \right\rangle \tag{11}$$

sourced by the entropy current $J^S_\mu = \partial_\mu S/T$. The continuum description of torsion adopted here is the sourced Proca equation for this vector potential,

$$\Box A_\mu - \partial_\mu(\partial \cdot A) = J^S_\mu, \qquad J^S_\mu \propto \frac{\partial_\mu S}{T}, \tag{12}$$

which reduces in the static limit to a Poisson equation sourced by the entropic charge density. This formulation, developed in the stability analysis of scalar knots, supplies a concrete field equation and a direct link to the numerically measured entropic charge Q.

The earlier characterization of torsion as an "emergent curl" is understood as a qualitative precursor to this equation. The map from discrete antisymmetric (directed) bond responses on the residual graph to the continuum Proca field remains a motivated coarse-graining; a fully controlled continuum limit of the antisymmetric sector is still lacking.

3.3. Projection to the Torsion Tensor

In the continuum effective field theory the vector potential enters the torsion tensor of Poincaré Gauge Theory through the antisymmetric contribution

$$T^{\lambda}{}_{\mu\nu} \supset A_{[\mu}\partial_{\nu]}S/T. \tag{13}$$

This is precisely the continuum projection of the lattice-level directed bond response. The full geometric language of torsion—connection, contorsion, Proca dynamics—is therefore an effective description, not a fundamental addition to the ontology. It is the natural mathematical tool for articulating, in the emergent $3+1$-dimensional chart, the consequences of an asymmetry that already exists among the residual bonds of the scalar lattice.

The construction is consistent with the continuum limit derived in Section 2: the same residual bonds that survive the spatial cutoff and carry the elevated coordination also carry the directed response that projects to A_μ. No new fields or parameters are required.

3.4. Relational Positivity and the Protection of Global Structure

The directed response remains strictly relative. Writing the effective affinity as

$$w_{ij}^{\text{eff}} = w_{ij}^{\text{sym}} + \delta_{ij}, \tag{14}$$

where $w_{ij}^{\text{sym}} > 0$ is the base (symmetric) affinity that survives the spatial cutoff and $\delta_{ij} = -\delta_{ji}$ is the antisymmetric correction, we impose the relational positivity condition

$$w_{ij}^{\text{sym}} > |\delta_{ij}| \qquad \forall\, i, j. \tag{15}$$

This single inequality guarantees that every effective graph distance stays strictly positive-definite. Local stretching is therefore permitted, but only up to the point at which it would invert the positive-definite structure maintained by the network as a whole. In this precise sense the prohibition on closed timelike curves and on configurations that would realize a grandfather paradox is relational rather than absolute: no finite collection of directed corrections can destroy the global causal structure that the remaining positive bonds collectively enforce.

3.5. Consistency with Residual-Bond Mediation

Because the directed component lives on residual primordial bonds, the effective torsion is automatically tied to the other phenomena already attributed to those bonds (sub-local entanglement and horizon entropy counting). This supplies a unified lattice origin for several continuum structures that had previously appeared as separate features of the effective field theory.

The same projection logic that converts the discrete Dirichlet energy into a continuum kinetic term and Einstein–Hilbert action therefore converts directed bond asymmetry into continuum torsion. In both cases a lower-dimensional or purely discrete feature is described using the richer language of the emergent geometry.

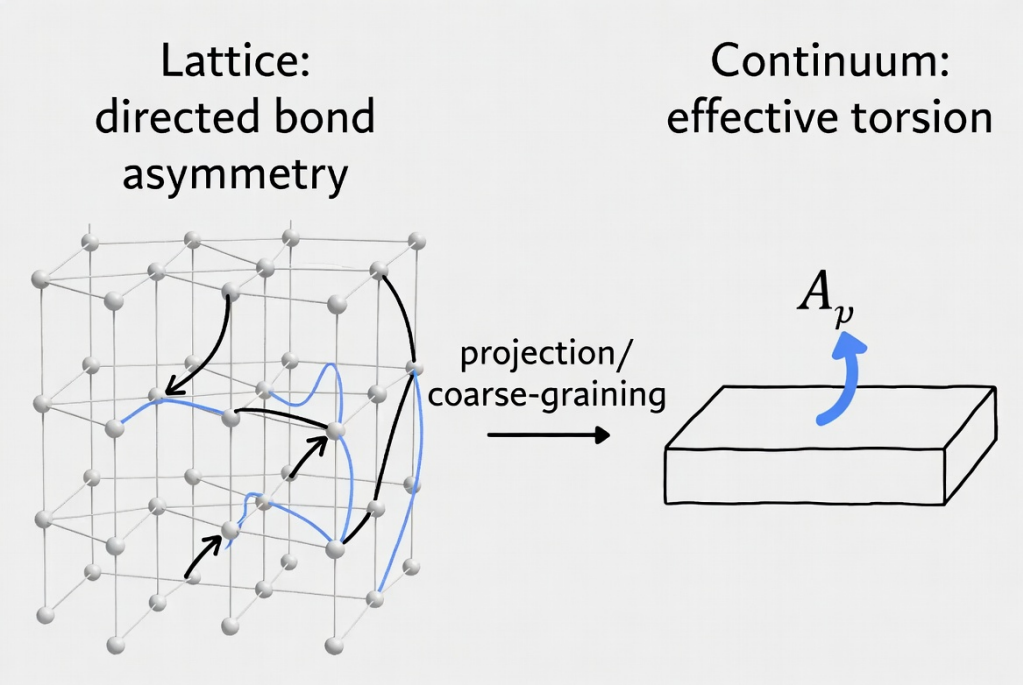


Figure 1: Lattice-level directed bond asymmetry (left) projects, after coarse-graining, to continuum effective torsion (right).

4. 0D Cores and Chart-Dependent Descriptions

Plain-language overview.
Black holes give the clearest example of the projection idea. From the outside we measure a certain radius and speak of an interior. In the underlying network, however, there is no extended interior. What sits at the center is a tiny, dimensionless knot made of the original strong links. The radius we measure is simply the distance at which enough surrounding links have been drawn into that knot. The continuum language of a horizon and an interior is our way of describing, from the outside, a structure that remains zero-dimensional in the original network. The same pattern—using higher-dimensional mathematics to talk about a lower-dimensional feature—appears again when we describe torsion.

4.1. Horizon as Chart-Dependent Partition

At the apparent or event horizon the exterior observer's emergent spatial chart is partitioned. Every bond w_{ij} whose graph distance $d_{\text{eff}}(i,j)$ would cross the surface $r = 2M$ in the exterior coordinate chart is excluded from that exterior chart: it is dropped from the exterior Dirichlet energy and from the induced metric $\gamma_{ab}^{\text{exterior}}$.

These bonds are called *partitioned*. The term does not mean that the bonds are destroyed. The underlying pre-geometric lattice remains all-to-all; every affinity w_{ij} continues to exist. Partitioning is a chart-dependent bookkeeping act performed when the exterior observer's effective spatial geometry is constructed. The same bonds that are omitted from the exterior description remain present in the full network and become part of the residual structure available to the interior. In particular, partitioning must not be understood as the physical breaking or

deletion of bonds; any such reading would be inconsistent with the pre-geometric ontology.

4.2. Interior as Planck-Scale 0D Core

Because the partitioned bonds are never removed from the lattice, the interior region reverts to the primordial, maximally overlapping structure — the same bare index set with all-to-all residual primordial bonds that defines the starting point of the ontology. No extended interior geometry exists. What remains is a fixed, local Planck-scale 0D core knot of residual bonds.

4.3. Observed Radius as Capture Radius

The Schwarzschild radius $r_s = 2M$ measured by an exterior observer is not the size of a larger interior object. It is the capture radius around that fixed 0D core: the distance at which local entropy gradients have drawn in enough surrounding primordial bonds to reach the bond count required by the area law

$$N_{\text{bonds}} = \frac{A}{4\ell_{\text{Pl}}^2}. \tag{16}$$

Larger mass simply means that more external bonds have been captured into the same local core. The core itself remains Planckian and dimensionally zero.

4.4. Information and Sub-Local Correlations

Information that appears lost to an exterior observer continues to be carried by the residual primordial bonds that cross the horizon partition. These bonds were never part of the emergent spatial graph; they remain sub-local. Unitarity is therefore preserved by the same mechanism already used for entanglement.

4.5. The Projection Parallel

The 0D-core construction and the treatment of torsion follow exactly the same logic. In both cases a feature that is fundamentally discrete or lower-dimensional in the pre-geometric lattice (a 0D knot of residual bonds; a directed asymmetry among those bonds) must be described, once the continuum limit is taken, with the richer mathematical language of the emergent geometry (a capture radius and horizon partition in $3+1$ dimensions; a vector potential and torsion tensor). The higher-dimensional description is a projection and a convenience; it does not elevate the underlying structure into an independent continuum object.

This precedent confirms that the framing developed for torsion in Section 3 is not an isolated adjustment. It is the consistent application of a pattern already required by the ontology.

5. Temporal Stacking and Directed Structures

Plain-language overview.
Time appears when the original network is stacked in successive layers. Each layer is a "now," and each now already possesses a spatial shape given by the geometry that has emerged on that layer. The layers are kept a positive distance apart by the same rule that keeps every link positive, so one now cannot collapse onto another and causal paradoxes cannot arise. The

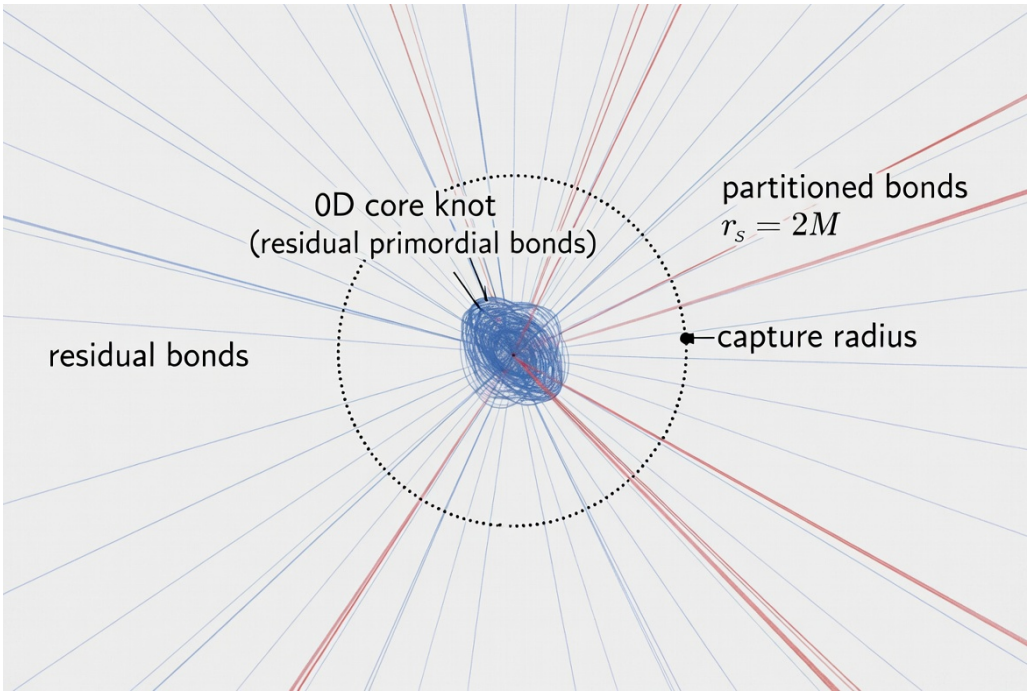


Figure 2: Schematic of a Planck-scale 0D core knot of residual primordial bonds. The dashed circle marks the capture radius $r_s = 2M$. Blue lines represent residual bonds that remain connected; red lines represent bonds partitioned (excluded) from the exterior observer's chart.

directed flow from past layers to future layers is simply the network's way of ordering the stack. When we describe all of this with continuum mathematics we obtain ordinary time, a finite speed of light, and a thermodynamic arrow. Those continuum notions are our language for patterns that already exist in the stacked network.

5.1. Stacking as an Additional Index

Temporal structure arises by stacking successive copies of the spatial lattice along a new discrete index n. Vertical bonds of the form

$$\Gamma_{\text{temp}} = \frac{1}{2} \sum_n \sum_i \frac{\lambda_\perp}{T_\perp} (S^i_{n+1} - S^i_n)^2 \tag{17}$$

are introduced. In the continuum limit along each world-line this yields the Cattaneo equation

$$\tau_C \partial_T^2 S + \partial_T S = \kappa_\perp \Delta S, \tag{18}$$

which enforces a finite signal speed $c = \sqrt{\kappa_\perp / \tau_C}$. The emergent four-metric is disformal,

$$ds^2 = \gamma_{ab}(x)\, dx^a dx^b - \frac{(\partial_T S\, d\tau)^2}{T_\perp}. \tag{19}$$

Proper time is the directed integral

$$\tau_S = \int \frac{\nabla S}{T} \cdot dx. \tag{20}$$

5.2. The Shape of "Now" and the Separation Between Successive Nows

In the discrete construction a "now" is not a single global simultaneous surface. It is a local observer chart (or a small atlas of overlapping charts) at a fixed stacking index n. The shape of that now is the emergent spatial geometry produced by Dirichlet minimization on that layer: the induced metric γ_{ab}, the local value of β_{eff}, the residual directed asymmetries, and the parallel-gradient offsets between neighboring charts.

Successive nows are separated by the vertical affinities. Because every vertical bond remains positive and is subject to the relational positivity condition of Section 3.4, the separation between adjacent layers is strictly positive-definite. In the continuum limit this separation becomes a directed proper-time interval along the stacking direction. One cannot collapse a future now onto a past now, nor create a closed loop in the stacking direction.

The shape of the present is therefore continuously informed by the immediately adjacent layers through the vertical coupling. Previous nows and future nows remain distinct, ordered, and positively separated.

5.3. Directedness of Temporal Flow

The vertical bonds are themselves subject to the same temperature-modulated, gradient-dependent affinities that generate the spatial directed response. Consequently the stacking direction inherits a preferred orientation: entropy production and the Onsager bilinear $\sigma \geq 0$ select a thermodynamic arrow. The continuum description of this arrow—the Cattaneo propagator, the disformal factor, and the directed proper-time integral—is again an effective projection. The underlying lattice supplies only scalar differences and positive affinities; the continuum language of a one-dimensional time with a definite direction is the convenient higher-dimensional description of those differences once the stacking limit is taken.

The arrow of time is treated throughout as an effective projection of the underlying scalar dynamics. Temperature-tuned stacking and the global entropy gradient supply a directed temporal order without additional postulates. The Two-State Vector Formalism, introduced in earlier work as a possible continuum description of time-symmetric correlations, is not required by the lattice ontology. Whether an effective retrocausal channel can be recovered inside the emergent temporal structure remains an open question; it is not assumed as a fundamental ingredient of the present framework.

5.4. Relational Consistency with Causal Structure

Because every vertical affinity remains positive and obeys the same relational positivity condition derived for the spatial bonds, the stacked geometry inherits global hyperbolicity. No continuous deformation generated by directed responses (spatial or temporal) can produce closed timelike curves. The arrow of time and the prohibition on causal paradoxes therefore share a common lattice origin: both are protected by the requirement that directed corrections never invert the positive-definite structure maintained by the network as a whole.

5.5. Projection Summary for Time

Temporal structure is not a fundamental extra dimension added to the ontology. It is the continuum projection of an ordered stacking of the original scalar line, with directedness arising from the same residual-bond asymmetries that project to torsion. The mathematical language of a one-dimensional time, successive "nows" that possess spatial shape, finite separation between those nows, a finite speed of light, and a thermodynamic arrow is the natural tool for describing, in the emergent chart, patterns that already exist among the stacked affinities.

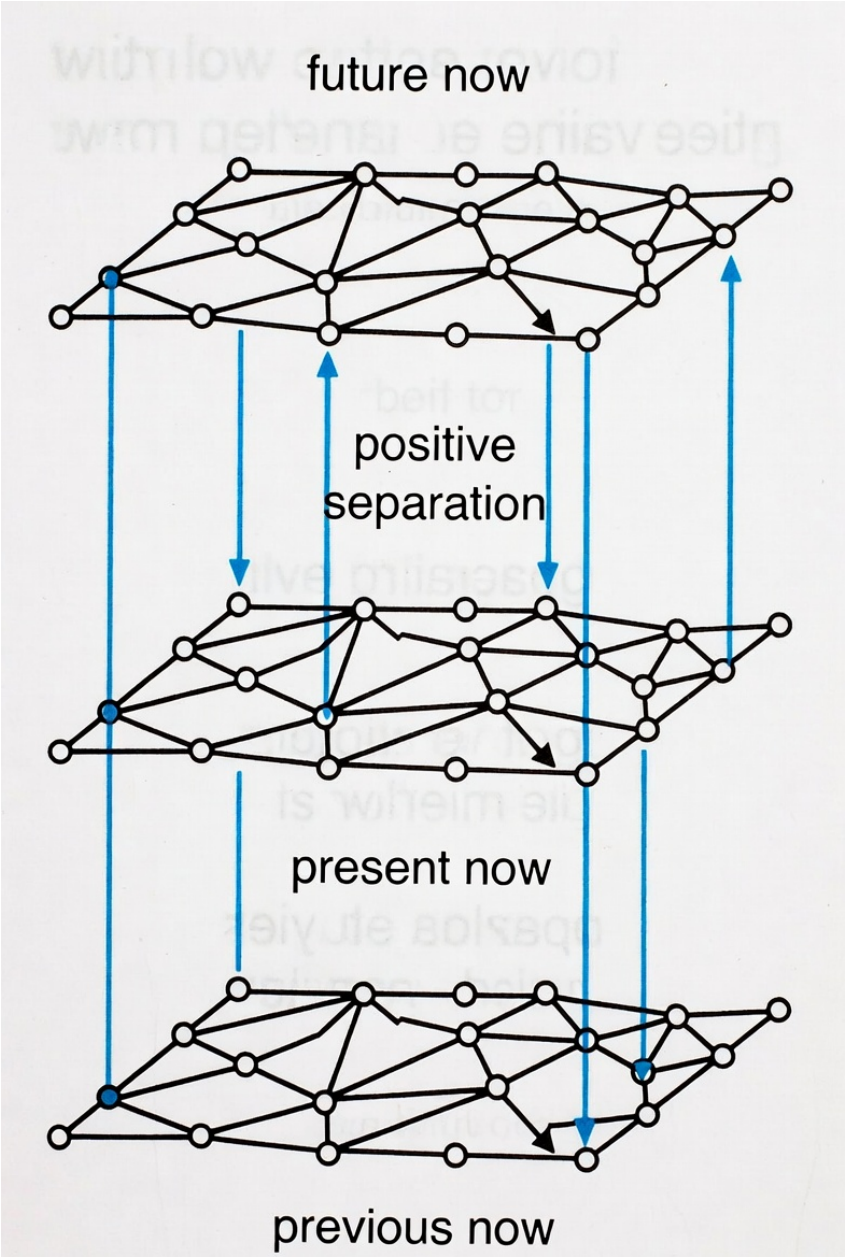


Figure 3: Successive stacked layers representing previous, present, and future "nows," connected by vertical bonds and maintained at positive separation.

6. Multiverse Fibers, Branching, and Overlapping Histories

Plain-language overview.
Once space and time have emerged from the original network of points and links, the same network can be stacked many times over. Each stacked copy is a possible history. Early on, when the system is hot, the copies are locked tightly together and remain identical. As the system cools they gradually unlock and become distinct branches. Branches that contain more disorder (higher entropy) simply occur in more ways, which automatically gives them higher probability—the Born rule. All of these branches still share one special point: the single coldest location in the entire network. That shared cold point is the geometric origin of the low-entropy beginning of the universe. Even after the branches have separated, a few of the original, very strong links continue to connect them, so the histories are never completely cut off from one another. The elaborate language of "many worlds" is simply a convenient way of describing this further stacking and residual linking of the original network.

6.1. Inter-Branch Coupling and Decoherence

One introduces $K \to \infty$ copies of the four-dimensional histories together with the inter-branch coupling

$$\Gamma_{\rm mv} = \frac{1}{2}\int d^4x\, \frac{\lambda_{\rm mv}}{T_{\rm mv}(\tau)}(S^{(k)} - S^{(k+1)})^2, \tag{21}$$

where the effective inter-branch temperature scales as $T_{\rm mv}(\tau) \propto 1/\langle T(\tau)\rangle$. At early times $\langle T\rangle$ is high, $T_{\rm mv}$ is low, the coupling is stiff, and all branches remain identical. As the universe cools, $T_{\rm mv}$ rises, the coupling softens, and the branches decohere. The magnitude of the decoupling is enormous ($\Delta\Gamma_{\rm mv} \sim 10^{104}$), corresponding to the total entropy of the observable universe.

6.2. Born Rule from Entropy Counting

Higher-entropy branches possess exponentially more lattice realizations. The natural measure on the fiber is therefore proportional to $\exp(S_{\rm total})$. In the continuum limit this measure reproduces the Born rule. No additional probability postulate is required; the weighting follows directly from the counting of micro-states of the underlying scalar lattice.

6.3. The Shared Coldest Vertex and the Past Hypothesis

A simple but powerful lemma follows from the form of the inter-branch coupling. In the global minimum of the full multiverse action, exactly one lattice site achieves the absolute minimum temperature $T_{\rm min} > 0$ and is shared by every macroscopic branch. Splitting branches at that coldest site would cost infinite action unless the scalar difference vanishes. Consequently every macroscopic history shares the identical coldest vertex. The low-entropy initial condition (the Past Hypothesis) is thereby realized geometrically: the fiber bundle narrows to a single shared point that is forced to be the coldest and narrowest locus of the entire structure.

6.4. Overlap via Residual Primordial Bonds

Although the branches decohere macroscopically, they continue to overlap at the level of residual primordial bonds. These sub-local connections, which never participated in the dimensional emergence of any single branch, persist across the fiber and mediate the same correlations already responsible for entanglement and horizon entropy counting. The branches are therefore not wholly separate universes; they remain linked by the original pre-geometric lattice at the residual-bond level.

6.5. Projection and Relational Protection

The continuum language of a many-worlds fiber, of decohering branches, of a Born-rule measure, and of a shared low-entropy origin is again an effective projection. The underlying ontology supplies only a single scalar line that is stacked and then fibered, together with positive affinities subject to the relational positivity condition of Section 3.4. That condition continues to protect the entire construction: directed corrections, whether spatial, temporal, or inter-branch, cannot invert the positive-definite structure of the network. Global hyperbolicity and the absence of closed causal curves are preserved across the fiber.

In this way the multiverse is not an additional ontological commitment. It is the continuum description of the further stacking and residual-bond linking of the original scalar lattice, with the shared coldest vertex supplying a geometric rather than an ad-hoc resolution of the Past Hypothesis.

7. The General Pattern of Effective Projections

Plain-language overview.
Everything discussed so far follows one simple pattern. The underlying reality is a network of points carrying entropy values and linked by temperature-sensitive connections. That network settles into arrangements that already possess a definite shape and structure. When we wish to describe that shape in a way our minds can grasp and calculate with, we use the language of mathematics—ordinary geometry, torsion, time, successive moments, and branching histories. Mathematics is not an extra ingredient added to the universe. It is simply the most precise language we have for articulating a geometry that is already there. The continuum structures we write down are not more fundamental than the network; they are our way of talking about the patterns the network produces. The present section gathers the individual cases into a single statement of this projection principle.

7.1. The Pattern Stated

The pre-geometric ontology consists solely of a countable index set, a scalar entropy density on each index, and temperature-modulated affinities. All further structure arises by free-energy minimization of the associated Dirichlet energy subject to the relational positivity condition that every effective affinity remain positive-definite.

Once the continuum limit is taken, the same ontology is described by the richer language of differential geometry and field theory. In every case examined in this paper the higher-dimensional

geometric object is an *effective projection*:

- the continuum kinetic term and Einstein–Hilbert action are the Γ-limit of the discrete Dirichlet energy;
- continuum torsion is the projection of directed (antisymmetric) bond responses (realized as a sourced Proca equation);
- black-hole interiors appear as chart-dependent partitions around fixed 0D core knots;
- temporal structure and the shape of successive "nows" are the projection of ordered stacking;
- the multiverse fiber, the Born rule, and the shared low-entropy origin are the projection of further stacking and residual-bond linking.

No new fundamental fields or dimensions are introduced. The continuum mathematics is the natural tool for articulating, in the emergent chart, consequences that are already present in the scalar lattice.

7.2. Relational Positivity as the Common Protector

Across all of these projections the same relational condition operates: the symmetric part of every affinity must dominate any directed correction. This single requirement guarantees that graph distances remain positive-definite, that the emergent geometry stays globally hyperbolic, and that no continuous deformation can produce closed timelike curves or grandfather-paradox configurations. Spatial geometry, torsion, temporal stacking, and the multiverse fiber are therefore protected by one and the same lattice-level principle.

7.3. Internal Consilience

Because every geometric structure is obtained by projection from the identical scalar line, the theory achieves a high degree of internal consilience. Phenomena that appear distinct in the continuum—gravity, torsion, the arrow of time, branching, and the Past Hypothesis—share a common microscopic origin in residual primordial bonds and Dirichlet minimization at the working value $\beta \simeq 3$. The continuum limit derived in Section 2 supplies the bridge that makes these identifications precise rather than merely analogical.

7.4. What Is Not Claimed

The projection principle does not assert that the continuum descriptions are unreal or optional. They remain the correct and useful language for observers who inhabit the emergent geometry. It asserts only that those descriptions are not fundamental additions to the ontology. The fundamental objects remain the scalar values and the positive, temperature-modulated affinities of the pre-geometric lattice. Mathematics is the necessary medium through which that lattice geometry becomes intelligible and calculable to us; it is not itself an extra layer of reality.

8. Discussion and Outlook

Plain-language overview.
This paper has not changed the predictions of Entropic Universe Theory. It has tried to make the foundations clearer. The universe, on this view, begins as a simple network of points and links. When that network settles, it already possesses geometry, time, and even branching histories. Mathematics is the precise language we use to describe those features so that we can calculate and test them.

Direct examination of the settled network shows a high-coordination, low-clustering geometry consistent with three-dimensional continuum behavior. Earlier attempts to fix the strength of gravity by a classical counting of links, and an intermediate residual-layer interpretation of the elevated coordination, have been withdrawn; the overall scale of the continuum curvature term remains an open internal question. The relational rule that keeps every link positive continues to protect the whole structure from causal paradoxes. With these clarifications in place, the existing experimental proposals remain the proper next step for testing the theory against nature.

8.1. Summary of Results

We have supplied a continuum limit of the discrete Dirichlet energy via Γ-convergence, recovering the continuum kinetic term and an emergent Einstein–Hilbert contribution whose overall scale is left open after withdrawal of the earlier classical isostatic matching and of the residual-layer interpretation, together with approximate diffeomorphism invariance. We have shown that continuum torsion arises as the projection of directed bond responses (realized via a sourced Proca equation) subject to a relational positivity condition that preserves global hyperbolicity. The same projection principle accounts for black-hole interiors as chart-dependent partitions around 0D core knots, for temporal structure and the shape of successive "nows," and for the multiverse fiber with its shared coldest vertex and Born-rule weighting. In every case the higher-dimensional geometric language is an effective description of patterns already present in the scalar lattice.

8.2. What Remains Unchanged

No equation, numerical result, or empirical prediction of the existing EUT framework has been altered. The laboratory tests proposed in the companion papers—especially the non-magnetic phononic and diamagnetic-levitation protocols—remain the primary route to empirical confrontation. The present work simply renders the conceptual and mathematical foundations more transparent and more internally consistent.

8.3. Mathematical Status and Next Steps

Two points of the continuum mathematics remain at different levels of completeness.

First, the continuum limit of the discrete Dirichlet energy is established by invoking existing Γ-convergence and spectral-convergence theorems for weighted graphs and random geometric graphs. The application to the temperature-modulated, post-minimization, cutoff-restricted weights of the present framework is consistent and correctly cited, yet a complete proof with

explicit error estimates tailored to these exact weights has not been written here. Such a derivation remains a natural next mathematical step.

Second, the identification of lattice-level directed (antisymmetric) bond responses with the continuum vector potential A_μ and the torsion term in Poincaré Gauge Theory is conceptually coherent, but it remains a motivated coarse-graining rather than a fully controlled continuum limit. A more explicit calculation tracking the antisymmetric component from the bond responses into the Proca structure would strengthen this identification.

A third point has been clarified by direct measurement of the surviving graph at the working value $\beta \simeq 3$. With a corrected weight formula and improved high-weight sampling, the mean coordination number is high ($\langle k \rangle \approx 30$), yet the graph exhibits low clustering ($C \approx 0.023$) and essentially zero combinatorial tetrahedra. The surviving edges are overwhelmingly strong (mean weight ≈ 0.94). These measurements show that earlier attempts to fix the continuum Einstein–Hilbert prefactor by inserting $\langle k \rangle$ into a classical isostatic (Maxwell) formula were misplaced. That matching is withdrawn. The overall scale of the curvature term must be expressed, if at all, in terms of quantities native to the scalar lattice; no such internal expression is claimed here.

The earlier residual-layer interpretation of the elevated coordination is also withdrawn on the basis of the same measurements: there is no significant population of weaker edges above the spatial cutoff. The structural picture that replaces it is a high-coordination, low-clustering network consistent with emergent three-dimensional continuum geometry.

In addition, $\beta \simeq 3$ is treated as a fixed working value. An explicit dynamical drive toward this value was present in earlier numerical implementations and has been removed. No native selection mechanism that stably produces $\beta \simeq 3$ under the pure Dirichlet + pruning dynamics has yet been demonstrated. Free-energy constructions that eliminate an external chemical-potential parameter can place a stationary point near the working value on frozen snapshots, but do not yield a stable dynamical attractor under continuous evolution.

None of these gaps or revisions affects the internal consistency of the projection principle, the relational positivity condition, or the continuum-limit statements for the kinetic term. They represent well-defined opportunities for further mathematical and numerical work rather than present contradictions.

Supporting numerical scripts that reproduce the structural results (clustering, triangle counts, weight histograms) are deposited with this version of the paper.

8.4. Closing Remark

The theory continues to rest on a single scalar line and temperature-modulated affinities. Everything else—space, torsion, time, branching, and the low-entropy past—is the geometry that this line produces, articulated in the mathematical language required for human comprehension and calculation. The task now is to test that geometry against experiment.

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